Wormhole teleportation with a Gaussian sparse SYK instance: circuits for IBM Nighthawk

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Abstract

In response to a question from the authors of [1], we give one N=8 Gaussian sparse SYK instance with K=24 terms for the traversable-wormhole teleportation protocol, with routed circuits for IBM Nighthawk; it passes every gate of the frozen verifier. Its routed circuits cost 216/225 two-qubit gates (FIX/SYM protocols), 23% and 5% *more* than their binary Eq. 11 compiled with the same pipeline (175/215), and 30% and 45% fewer than a typical K=24 draw (309/410). In a FakeNighthawk noise simulation the FIX circuit still keeps 25 to 35% more teleportation signal than Eq. 11 over three routing-seed lists, and 18% more when the term order is optimized on both sides; this is our robust claim. SYM keeps 25 to 29% more in canonical term order but is at best a tie when the term order is optimized. Against Eq. 11 in the term orders its authors used, the map-S2 margins are 23 to 50% (FIX) and 51 to 67% (SYM); in map S3, SYM ties. Nothing has run on hardware.

Problem. Two coupled Sachdev–Ye–Kitaev (SYK) models in a thermofield double (TFD) state model a traversable wormhole: a qubit swapped into the left side at $-t_0$ reappears on the right at t_1 when the left-right coupling $e^{i\mu V}$ has $\mu < 0$ [2, 3, 4]; small instances have run on hardware [5], with debate on how well they scramble [6]. Sparse SYK keeps K of the $\binom{N}{4} = 70$ four-Majorana terms and stays chaotic above a threshold K [7, 8, 9]. Ref. [1] measures the signal $\Delta I = I_{PT}(\mu{=}{-}12) - I_{PT}(\mu{=}{+}12)$, with I_{PT} the base-2 mutual information between a reference qubit P and the readout qubit T, at $\beta = 3$ on 10 qubits, in the symmetric (SYM, $t_0 = t_1$) and fixed-injection-time (FIX, $t_0 = 1.8$) protocols, and ran a binary K=10 instance (their Eq. 11) on IBM hardware. They asked us for a Gaussian one ($\mathrm{Var}\, J = 1.640625/K$, $K \approx 24$) with large ΔI in both protocols and few two-qubit (2q) gates after routing on IBM Nighthawk (our bar: $\leq 0.70\times$ the typical median, FIX 216.3, SYM 287.0).

Result. An *instance* is one term set plus one Gaussian coupling draw. Table 1 lists the three instances that passed every gate, plus two controls; Appendix A prints the recommended instance, 19277_201. The first two share a term set. The third, 1718_714, passes by the pre-registered rule on seed list A (FIX +2.8%, SYM +0.8%) but not on held-out lists B and C (FIX −0.05%, −3.6%; SYM −1.4%, −3.1%): borderline. 19277_201 keeps +25–35% / +25–29% more noisy ΔI than Eq. 11 and 8.1 / 9.4 times a typical draw. The funnel: 32 instances on 16 term sets passed the cost, chaos, noiseless-signal and TFD-fit checks; 23 on 14 sets also passed robustness (cost on three seed lists; eight noiseless ΔI peaks $\geq 1.02\times$ Eq. 11's; ≥ 45 of 50 draws with 1% coupling noise keep every peak above Eq. 11's); 3 on 2 sets passed the noise test. The code repository ships 16 instances: 14 of the 32 that passed (including the three in Table 1) and 2 that failed only the SYM cost bar on list B (290 and 291 against 287.0). The cheapest instance that passed (FIX 197, K=25) fails the noise test.

Chaos. 19277_201 is consistent with the K=24 ensemble: level-spacing ratio $r = 0.44$ / 0.45 in the two parity sectors (26th / 28th percentile; Poisson 0.386, GOE 0.53) and spectral form factor (SFF) at the 65th percentile, both within the bands; its out-of-time-order correlator (OTOC) decays faster than 91% of typical draws (decay time 2.94 against our bar 2.95). With 8 levels per sector, r is weak evidence. The time-averaged SFF lies inside the K=24 band (5th–95th percentile of 2000 draws) in 18 of 18 time windows (155_129: 17 of 18), with log-log ramp slope 1.08 (Eq. 11: 0.53); the size-winding phase reverses after the kick for all 8 insertions in all three Hamiltonians; the two-point function's mean half-decay time is 2.08 (Eq. 11: 1.96), with late $|G|$ on $[8, 20]$ of 0.072 (0.069). A selected sparse instance can carry a signal without the chaos behind the protocol (Sec. 5 of [9]; [6]), so we also checked mirror symmetry: no exact $E \to -E$ map exists, and the near-mirror weight $m(H) = 0.217$ sits at the 98th percentile of the ensemble (farther from a mirror than 98% of draws).

Selection. 19277_201 is about 1 in 1.5×10^{13} draws from a defined pool (uniform K=24 term sets with i.i.d. Gaussian couplings: 5×10^8 term-set draws for the 10 sets whose couplings we screened, times 7.0×10^7 coupling draws for 231 screen passes), so its exact pair-(1,2) ΔI peak (0.39 / 0.30) is 2.7 / 2.4 times the typical median (0.15 / 0.12). The "8 to 9 times typical" is at circuit level, where Trotter error, routed cost and selection compound.

New beyond [1]. They already report Gaussian K_c=24 I_{PT} curves (Fig. S15) and a one-step K=24 cost of 81 2q gates (Table S1). Our addition is one explicit Gaussian K=24 instance with routed circuits predicted to keep more FIX signal than Eq. 11 under the same noise model.

Method. Term sets: 5×10^8 uniform K=24 term-set draws ranked by a cheap predicted cost, 30,842 kept, the top 200 routed on FakeNighthawk. Couplings: up to 2×10^7 i.i.d. Gaussian draws per set, screened on chaos, the eight ΔI peaks and Trotter error; the couplings are selected draws, never continuously optimized. Circuits: one Lie–Trotter step (FIX) or Suzuki-2 step (SYM), the Jordan–Wigner map S2 of [1] for both protocols, canonical greedy term order, per-block pytket 2.18.5 GreedyPauliSimp and FullPeepholeOptimise, then Qiskit 2.5.2 at optimization level 3 on FakeNighthawk; readout times, TFD fit and Trotter error in Appendix B. Noise: Aer density-matrix simulation from FakeNighthawk (qiskit-ibm-runtime 0.49.0) on the shipped circuits, two-qubit tomography of P and T.

Instance	K	t_1	2q	noiseless ΔI	noisy ΔI FIX	noisy ΔI SYM	vs Eq. 11	× typical
19277_201 (set 1)	24	2.6 / 2.35	216 / 225	0.33 / 0.28	0.116 ± 0.005 [0.120]	0.096 ± 0.0006 [0.094]	+25–35% / +25–29%	8.1 / 9.4
155_129 (set 1)	24	2.45 / 2.15	216 / 225	0.27 / 0.25	0.097 ± 0.004 [0.102]	0.087 ± 0.0007 [0.085]	+6–14% / +14–18%	6.8 / 8.5
1718_714 (set 2)[b]	24	2.55 / 2.5	208 / 250	0.25 / 0.26	0.092 ± 0.002 [0.094]	0.075 ± 0.0003 [0.075]	−4 to +3% / −3 to +1%	6.4 / 7.3
Eq. 11 (binary)	10	2.35 / 2.05	175 / 215	0.21 / 0.20	0.089 ± 0.0009 [0.083]	0.074 ± 0.001 [0.074]	ref.	6.3 / 7.2
Typical K=24 (n=20)	24	–	309 / 410	0.060 / 0.079[a]	0.014	0.010	−84% / −86%	1 / 1

Table 1: Instances that pass every gate, and controls. Pairs are FIX / SYM. Sets 1 and 2 are term sets d6a76272 and 2f2d8f4f in the repository. t_1: readout time ($t_0 = 1.8$ for FIX, $t_0 = t_1$ for SYM). 2q: routed cz count, the lower median over 20 routing seeds, also the shipped circuit's count. Noiseless ΔI: routed circuit with the fitted TFD at t_1, bits. Noisy ΔI: FakeNighthawk gate and readout noise, median over the central 5 of 20 routing seeds (sorted by 2q, depth, seed) of seed list A; ± is half the range of the medians over seed lists A, B, C; [ ] is the shipped circuit alone. "vs Eq. 11": range over lists A, B, C, each against Eq. 11 on the same list; ratios use unrounded medians. "× typical": list A. Typical: median of 20 uniform K=24 draws, list A; [a]over the 17 draws with all 5 seeds; 2q is the 200-draw median. [b]borderline.

Error bars and the hardware run. Across the five central routing seeds of list A, 19277_201 ranges 0.106–0.121 (FIX) and 0.091–0.099 (SYM); every seed is above Eq. 11's median, and on held-out list B every seed beats every Eq. 11 seed. A single shipped circuit (Table 1, [ ]) moves by up to about 8% from the median. Our gate + readout noise has no idle decoherence (close to ideal dynamical decoupling, DD); adding idle T_1/T_2 is close to no DD. Ref. [1] ran with XpXm DD and measurement twirling, so the device should sit between the two. With idle noise, 19277_201 keeps +21% / +22% (0.078 / 0.066 vs 0.064 / 0.054), and 155_129 ties Eq. 11 on FIX (0.0642 vs 0.0644). The expected device gap to Eq. 11 is therefore about 0.01–0.03 bits. Run plan: 2 signs × 9 tomography settings × 14,900 (FIX) or 18,000 (SYM) shots, about 270k (FIX) and 320k (SYM) shots per protocol per Hamiltonian, for a standard error of 0.005 bits on ΔI (the FIX gap of 0.027 bits is then about 3.8σ); interleave Eq. 11 in the same session; a typical-draw control is optional.

Verification. Eq. 11 gives $r = 0.537163$ / 0.478812 against the printed 0.537 / 0.479; Eq. 11's exact pair-(1,2) ΔI peaks, 0.2127 (SYM) and 0.2765 (FIX), match Fig. S1 of [1] (0.212 and 0.276, read to ±0.003). Their Table S1 transition points: Appendix B. Only verdicts of a frozen verifier (checks and bars fixed before scoring) count; seed list A searched, B certified, C held out from the search. A blind re-derivation from the shipped files alone reproduced every number needed to run the shipped circuits to 2.8×10^{-7} (two text errors found and fixed, no number errors). We assume FakeNighthawk (revision 1) has the coupling map of the Nighthawk r2 device.

Limits. (i) FakeNighthawk is a simulated proxy for the device. With every FakeNighthawk error rate scaled up, 155_129 FIX falls to Eq. 11 at about 1.5 times the rates, while 19277_201 stays 16–25% above it up to 1.75 times. (ii) With term order optimized on both sides, Eq. 11 SYM routes to 134–135 2q; 19277_201 FIX keeps +18%, but SYM loses in two of three order searches (−10%; −10% / −11% on lists A / B) and ties in the third (+1% / −7%). (iii) Thin margins: 19277_201 FIX costs 216 against the bar 216.3 (list C: 217 against 219.1); its OTOC decay time is 2.94 against 2.95 and F 0.965 against 0.96; the second robust instance came from a search on the set that had just passed; the final search sent 6 instances on 5 sets to the verifier and only the borderline one passed, by +0.8% on SYM, within the 8–14% per-seed spread. (iv) After the first version of this note, the authors of [1] sent the Eq. 11 term orders they used (private communication). Against them the map-S2 margins are +23 to +50% (FIX, the backward convention worse for us) and +51 to +67% (SYM); in map S3 SYM only ties, so our SYM claim is for S2 only (Appendix B). (v) Ref. [1] printed 150 2q for Eq. 11 FIX in map S2; we get 175 (greedy order) to 195–206 (their orders), a gap we cannot explain (SYM in S3: 282 printed, 241–274 ours).

Code. `https://github.com/oharalabsai/syk-gaussian-n8`: `uv run verify` re-derives Eq. 11's known answers and every instance's 2q and noiseless numbers in about a minute; `uv run noise <instance>` gives its noisy ΔI.

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AI contribution. The authors are responsible for all content. An AI agent harness ran the search, simulations and first drafts under human direction; every number comes from a frozen verifier or a logged job. No AI is an author.

References

[1] Moongul Byun, Hyeonsoo Lee, and Keun-Young Kim. Quantum simulation of traversable-wormhole-inspired quantum teleportation in a chaotic binary sparse SYK model. *arXiv:2604.10090*, 2026.

[2] Ping Gao, Daniel Louis Jafferis, and Aron C. Wall. Traversable wormholes via a double trace deformation. *J. High Energy Phys.*, 2017(12):151, 2017.

[3] Juan Maldacena, Douglas Stanford, and Zhenbin Yang. Diving into traversable wormholes. *Fortschr. Phys.*, 65(5):1700034, 2017.

[4] Adam R. Brown, Hrant Gharibyan, Stefan Leichenauer, Henry W. Lin, Sepehr Nezami, Grant Salton, Leonard Susskind, Brian Swingle, and Michael Walter. Quantum gravity in the lab. I. Teleportation by size and traversable wormholes. *PRX Quantum*, 4:010320, 2023.

[5] Daniel Jafferis, Alexander Zlokapa, Joseph D. Lykken, David K. Kolchmeyer, Samantha I. Davis, Nikolai Lauk, Hartmut Neven, and Maria Spiropulu. Traversable wormhole dynamics on a quantum processor. *Nature*, 612:51–55, 2022.

[6] Bryce Kobrin, Thomas Schuster, and Norman Y. Yao. Comment on "Traversable wormhole dynamics on a quantum processor". *arXiv:2302.07897*, 2023.

[7] Shenglong Xu, Leonard Susskind, Yuan Su, and Brian Swingle. A sparse model of quantum holography. *arXiv:2008.02303*, 2020.

[8] Antonio M. García-García, Yiyang Jia, Dario Rosa, and Jacobus J. M. Verbaarschot. Sparse Sachdev-Ye-Kitaev model, quantum chaos, and gravity duals. *Phys. Rev. D*, 103:106002, 2021.

[9] Patrick Orman, Hrant Gharibyan, and John Preskill. Quantum chaos in the sparse SYK model. *J. High Energy Phys.*, 2025(2):173, 2025.

Appendix A: the recommended instance 19277_201. $H_a = \sum_{T=1}^{24} J_T\, \psi_a^i \psi_a^j \psi_a^k \psi_a^l$ with $T = (i\,j\,k\,l)$, $a = L, R$, the same convention as Eq. 11 of [1] ($\{\psi^i, \psi^j\} = \delta_{ij}$, ordered product, the factor $i^{q/2}$ absorbed in J_T), J_T drawn i.i.d. from $\mathcal{N}(0, 1.640625/24)$.

T	$(a\,b\,c\,d)$	J_T	T	$(a\,b\,c\,d)$	J_T	T	$(a\,b\,c\,d)$	J_T
1	1 2 4 7	+0.054155	9	1 4 5 8	+0.354397	17	2 4 5 6	+0.619194
2	1 2 5 7	−0.434389	10	1 4 6 7	+0.275300	18	2 4 6 7	+0.008317
3	1 2 6 7	+0.160388	11	2 3 4 5	+0.032474	19	2 5 6 8	−0.092359
4	1 2 6 8	+0.333668	12	2 3 4 7	+0.344660	20	2 5 7 8	−0.118641
5	1 3 4 5	+0.078696	13	2 3 4 8	+0.218100	21	2 6 7 8	+0.215055
6	1 3 5 6	−0.314504	14	2 3 5 7	+0.380139	22	3 4 5 7	+0.330082
7	1 3 5 8	−0.105945	15	2 3 6 7	+0.336518	23	3 4 5 8	−0.102157
8	1 3 6 7	−0.019269	16	2 3 6 8	+0.111073	24	5 6 7 8	+0.207052

Trotter term order (both protocols, both evolution stages): 1 4 8 22 2 5 10 12 17 21 3 6 14 16 23 24 7 11 18 19 9 13 15 20 (the greedy commuting order of the sorted term list). Times: FIX $t_0 = 1.8$, $t_1 = 2.6$; SYM $t_0 = t_1 = 2.35$; $\beta = 3$, $\mu = \mp 12$. Shipped circuits (routing seed, 2q): FIX 1019, 216; SYM 1018, 225. The VQA angles, layouts and QASM files are in the repository.

Appendix B: conventions and further checks. Readout time: t_1 is the argmax of the exact $I_{PT}(\mu{=}-12)$ on $[1, 4]$ (grid 0.05); FIX keeps $t_0 = 1.8$, SYM uses $t_0 = t_1$ (values in Table 1). Scanning t_1 by ± 0.2 on the shipped circuits, both instances beat Eq. 11 at every matched offset (worst case 1.21× FIX, 1.14× SYM, 155_129 at its t_1 against Eq. 11's best point), and FIX ΔI rises about 2% at +0.2 for all three, since t_1 maximizes I_{PT}, not ΔI. TFD: a 6-layer, 96-angle variational circuit applied to the infinite-temperature state $|I\rangle$, best of 720 random starts; fidelity $F = 0.965$ for 19277_201 (Eq. 11: 0.98, same procedure; our bar $F \geq 0.96$). With FIX t_0 set to the SYM argmax instead (2.35 / 2.15 / 2.05 for 19277_201 / 155_129 / Eq. 11), the exact FIX pair-(1,2) peak drops 3–4% for all three (19277_201 0.379 vs 0.395, Eq. 11 0.267 vs 0.276) and the ratio to Eq. 11 is unchanged (1.42 vs 1.43). Trotter: the t_0 stage evolves H_L only, and backward evolution reuses the forward term order. The Trotter error $\eta = |\max \Delta I_{\text{exact}} - \max \Delta I_{\text{Trotter}}| / \max \Delta I_{\text{exact}}$ on $[1, 4]$ with the fitted TFD is 0.094 (FIX) / 0.0017 (SYM) for 19277_201, against 0.162 / 0.012 for Eq. 11 (0.168 / 0.009 with the exact TFD). Eq. 11's noiseless Trotter peaks (FIX / SYM) are 0.2301 / 0.2147 under a term order and an H_L-only first stage that we reconstructed; the literal reading of [1] gives 0.2246 / 0.2130. Table 1 uses the reconstruction on both sides. The authors of [1] later confirmed the H_L-only first stage and sent their term orders (M. Byun, private communication, 23 October 2026), which differ between H_L and H_R (FIX H_L 8 5 6 1 9 3 4 2 7 0, H_R 8 5 6 4 9 3 1 7 2 0; SYM H_L 3 8 7 5 0 1 2 9 6 4, H_R 3 7 8 1 2 5 6 0 4 9; positions in Eq. 11 from 0, first applied first). In those orders the peaks are 0.2436 (FIX, backward step in the same order; 0.2239 reversed) / 0.2123 (SYM). Routed and noisy in map S2 (lists A, B, C), Eq. 11 in their orders takes 195–206 (FIX, both backward conventions) / 254–257 (SYM) 2q and keeps 0.081–0.090 (FIX, same order), 0.076–0.081 (FIX, reversed) and 0.057–0.063 (SYM); 19277_201 keeps +23 to +50% (FIX, the worse backward convention for us) and +51 to +67% (SYM) more. Against them the run plan's FIX gap is 2.9 to 4.3σ, and Eq. 11's Trotter error η (exact TFD) is 0.119 (FIX, same order; 0.190 reversed) / 0.002 (SYM). Map S3 on both sides: 19277_201 +30% / +27%, 155_129 +19% / +15% over Eq. 11 in greedy order (FakeNighthawk gate + readout, list A). Against Eq. 11 in their orders, the map-S3 SYM margin is +8, +5, −2% (lists A, B, C) for 19277_201 and −2, −5, −9% for 155_129: a tie at best. Transition points: with 10,000 realizations per K by their SI Sec. S2 method, the Gaussian points of their Table S1 come out at $K_1 = 22$ [21, 24] against 23 and $K_2 = 20$ against 23 (weakly constrained, [18, 69]), both Gaussian points inside our 95% intervals; binary $K_1/K_2 = 6$ [6, 7] / 9 [9, 12] against 6/8.